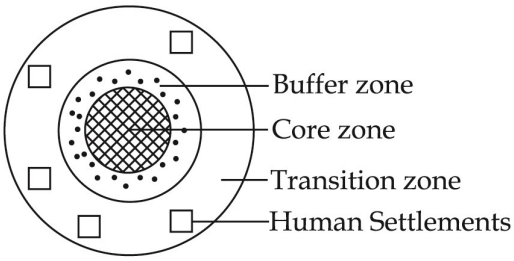
MARKING SCHEME

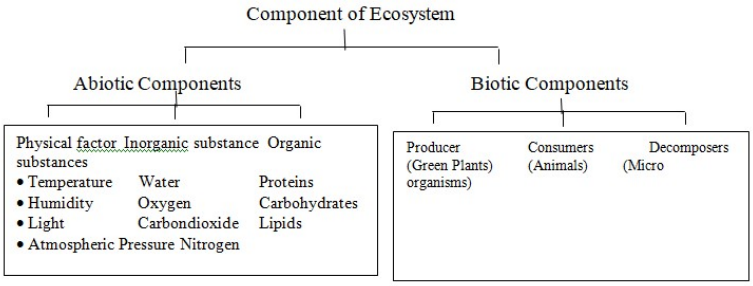
SUBJECT : 333 Environmental Science **MAX. MARKS : 8**

Q. NO.	EXPECTED VALUE POINTS FOR EACH STEP	DISTRIBUTION OF MARKS	TOTAL MARKS
SET	**Environmental Science** **OB1 Core Module**		
A			
1	(A)	1	1
2	(A)	1	1
3	(A)	1	1
4	(C)	1	1
5	(A)	1	1
6	(A)	1	1
7	(C)	1	1
8	(A)	1	1
9	(C)	1	1
10	(C)	1	1
11	(A)	1	1
12	(A)	1	1
13	(D)	1	1
	Optional Module 8A		
14	(C)	1	1
15	(D)	1	1
16	(B)	1	1
	Optional Module 8B		
14	(A)	1	1
15	(A)	1	1
16	(C)	1	1
	OB2 Core Module		
17	(a) Wealth in the form of new crops and medicinal plants. (b) Increased human resource consumption and pollution, over exploitation. Or any other (any two)	1 $\frac{1}{2}+\frac{1}{2}=1$	2

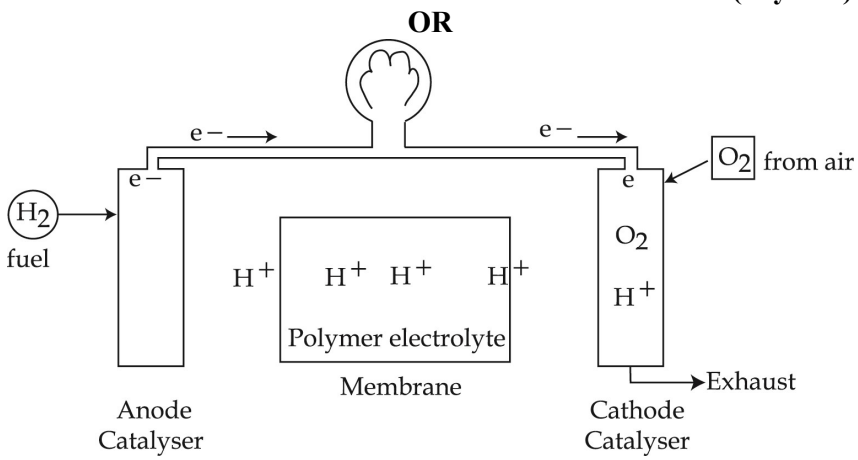
18	(a) Risk assessment, environmental management, post product monitoring **(any two)** (b) Reserved forests assigned to a village community. (c) Soil cannot sustain continuous cropping with high yielding single crop become the nutrients get exhausted; So sowing leguminous crops as a rotational crop is useful due to their nitrogen fixing ability. (d) Ganga Action Plan ; 1993 (e) Chlorine atoms from CFCs react with ozone to form chlorine monoxide and O_2 molecule which can further react to release more chlorine atoms. $Cl\square O_3 \rightarrow ClO\ \square O_2$ $2ClO\square O_2 \rightarrow 2Cl\square 2O_2$ (f) One horned rhinoceros, lions in Gir Forest.	$\frac{1}{2}+\frac{1}{2}=1$ **1** $\frac{1}{2}+\frac{1}{2}=1$ $\frac{1}{2}+\frac{1}{2}=1$ $\frac{1}{2}+\frac{1}{2}=1$ $\frac{1}{2}+\frac{1}{2}=1$	**1x6=6**
19	(a) EIA (Environment Impact Assessment) (b) Ramsar / Wetland convention (c) Poverty (d) Taxonomic / phylogenic diversity (e) Deciduous (f) Lead	**1** **1** **1** **1** **1** **1**	**1x6=6**
20	(a) Developer (b) Energy (c) Terrestrial (d) Pathogen	**1** **1** **1** **1**	**1x4=4**
21	(a) T (b) F (c) T	**1x3**	**3**
	Optional Module 8A		
22	Causes skin disorder/cancer	**1**	**1**
23	By collecting water on roof tops and directing it into storage tanks. By specifying areas where water can seep into the soil.	$\frac{1}{2}+\frac{1}{2}$	**1**
24	(i) Makes up for shortage of water supply (ii) Reduces water wastage (iii) Reduces cost **(any two)**	$\frac{1}{2}+\frac{1}{2}$	**1**

	Optional Module 8B		
22	Energy from tides is harnessed as they flow in and out Mean tidal range must be > 5m.	½ ½	**1**
23	Ethanol, Methanol, Biodiesel **(any two)**	$\frac{1}{2}+\frac{1}{2}$	**1**
24	(i) Using a cooking pan slightly larger than cooking plate/burner. (ii) Keep saucepan lids on. **(or any other) any two**	½ ½	**1**
	SA Core module		
25	a-iii, b-iv, c-ii, d-i	$\frac{1}{2}$ **x 4**	**2**
26	• Integrated Pest Management which avoids harmful chemical pesticides and use biological methods to get rid of the pests. • To keep the crop damage by pests to economically tolerable level. **OR** **Benefit -** Enhanced taste and quality of crops. **(or any other)** **Controversy** – Potential human health impact allergens from few of antibiotic resistance makers. **(or any other)**	**1** **1** **1** **1**	**2** **2**
27	India has high solar insulation; and is densely populated. So solar power is good for India. It is done by Ministry of New and Renewable Energy (MNRE) and Indian Renewable Energy Development Agency (IREDA).	½+½ **1**	**2**
28	Regions of the world that are very rich in biodiversity. Criterians (i) The area should support >1500 endemic species. (ii) They must have lost over 70% of the original habitat.	**1** ½+½	**2**
29	Factors – Improved agricultural practices, progress in medicine, better diagnosis and treatment of diseases.**(any two)**	**1 + 1**	**2**
30	By providing space the live; shelter/roof over their heads; providing food and water; and sunlight for the cows.	$\frac{1}{2}$ **x 4**	**2**
31	<table><tr><td></td><td>Rural settlement</td><td>Urban settlement</td></tr><tr><td>Occupation</td><td>Agriculture</td><td>Industrial workers, traders, professionals etc.</td></tr><tr><td>Social Interactions</td><td>Primary Contacts</td><td>Numerous contacts</td></tr></table>**OR** Depletion of natural resources, salination of soil, soil erosion, loss of mineral nutrients. **(any two)**	**1** **1** **1 + 1**	**2**
32	**Definition** –Cutting of trees including repeated lapping, felling and removal of forest-litter, browsing, grazing and trampling of seedlings. **Consequences –**	**1**	**2**

	Soil erosion increase in flash floods, loss of biodiversity. **or any other (Any two)** **OR** **Desertification -** The destruction of biological potential of the land which can ultimately lead to desert like conditions. **Activities –** Over exploitation of grazing land, excessive mining, indiscriminate cutting of forests, exploitation of water resources. **(or any other)**	$\frac{1}{2}+\frac{1}{2}$ **1** $\frac{1}{2}+\frac{1}{2}$	
	LA1		
33	Biosphere reserves are representative parts of natural and cultural landscapes extending over large areas of terrestrial or coastal/marine ecosystems which are Internationally recognized within UNESCO's Man and the Biosphere programme. • Three zones : (a) Core zone – inner most and protected. (b) Buffer zone – surrounds the core zone. (c) Transitional zone – outer most zone.  **OR** (i) Botanical Gardens – propagation of important threatened plant species. (ii) Gene Bank – preservation of seeds. (iii) Cryopreservation – storage of material at ultra low temperature. (iv) Conservation at molecular level – germplasm conservation, cloned DNA. **(any two with brief description)**	**1** **3** **1+1x2**	**4** **4**

34	The study of various aspects of a population is called demography. Birth rate – Rate at which births occur in a population. Birth rate $b = \frac{B}{N}$ Death rate – Rate at which deaths occur in a population. Death rate $d = \frac{D}{N}$ Growth rate is the result of the number of births minus the number of deaths per unit time divided by the total number of individuals in the population. Growth rate $g = \frac{(B-D)}{N} = \frac{G}{N}$	**1** **1** **1** **1**	**4**
	LA2		
35	Ecosystem is a functional unit of nature encompassing complex interactions between its biotic and abiotic components. **Pond** – **Abiotic** - Light, inorganic substances, organic substances **(explain each component)** **Biotic** – Autotrophs, Heterotrophs, Decomposers **(explain every component)**  	**1+2+2**	**5**
36	• Presence of any solid, liquid or gaseous substance including noise and radioactive radiation in the atmosphere in such concentration that may be directly and indirectly injurious to humans or other living organisms or interferes with the normal environmental process. • Particulate air pollutants **Source** / **Effect** SPM/dust — Smoke — Reduces visibility, asthma, Pneumoconiosis Flyash — Smoke from industrial chimneys — Settles down on vegetation Leachates are harmful • GaseousPollutants CO, CO_2 — Burning of wood and coal — Green house effect and respiratory diseases. SO_2 and H_2S — Power plants Refineries — Loss of chlorophyll in plants, Acid rain NO, N_2O — Motor vehicle exhausts — Irritation in eyes	**1** **1** **1**	**5**

	Hydrocarbons — Automobile and Petroleum industries — Cancer causing Respiratory problems **NOTE:** **Details of any one of particulate air pollutants and gaseous pollutant.** **Prevention and control –** - Proper ventilation of houses. - Using cleaner fuels. - Installing devices like filter and precipitators. - Installing high chimneys. **(any other four)** **OR** (a) CFC's deplete O_3 layer by splitting Ozone molecules to form oxygen. This depletes ozone layer over Arctic and Antarctic region and is called ozone hole. (b) Increase in temperature due to presence of CO_2, methane, nitrogen oxides is known as greenhouse effect. (c) Form of energy travelling through space. (d) Increase in temperature of cooling water by waste water from heaters and boilers. (e) Addition of substances which adversely affect the quality of soil or its fertility.	**2** **1** **1** **1** **1** **1**	**5**
	Optional Module 8A		
37	(a) Contamination of water by disease causing micro organisms. (b) Enrichment of nutrients that promotes excess growth of algae in water bodies.	**1** **1**	**2**
38	**Check for :** Leaks and drips ; Installation of water saving devices in toilets ; Open garden hoses, taps that are left on during brushing and washing of dishes. (Any four) **OR** Drought, expanding population, increasing irrigation demands, pollution.	**½x4** **½x4**	**2**
39	In Gadhigram (Kutch district) villagers contributed voluntary labors to build a check dam on a seasonal river to overcome water shortage. **(any other example)** **OR** -Use only the amount actually needed -See that there are no leaks in pipes/tanks.	**2** **½x4**	**2**

	-Do not leave tap open when not in use. -Use water from a bucket when washing a car. -Use minimum quantity of water. **(or any other four methods)**		
	Optional Module 8B		
37	Non-renewable -They are a finite energy resource once consumed lost forever. -Coal/Petrol Renewable -These sources which can be harnessed without depletion. -Solar/Wind	**1** **1**	**2**
38	On-site noise, emission of gases, air pollution **(any two)** **OR**  Working of H_2 fuel cell	**1 + 1** **2**	**2**
39	-Installation of photoelectric controls for outdoor lighting. -Switching off water pumps during non-peak hours. **(or any other)** **OR** -Compressed earth blocks. -Ferro-cemented roof channels -Solar Chimneys -Wind pump -Water solar PV pump -Wind generators -Decentralized waste water system **(or any other)**	**1 + 1** **½x4**	**2** **2**
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